WIC AI INNOVATION CHALLENGE COMPETITION

Knowledge Wave

AI-Powered Note-Taking for Microsoft Learn

An adaptive agent system that simplifies work and learning for diverse teams and neurodiverse users

THE CHALLENGE

Microsoft Learn

Microsoft Learn delivers hundreds of free training courses across 23 languages, yet suffers from a critical gap:

No integrated note-taking capability.

Users cannot annotate content, download materials for offline study, or capture insights without disrupting their learning flow.

Critical Pain Points

1. **Low Engagement**
   Fragmented workflows break concentration and reduce knowledge retention

2. **Poor UX**
   Context switching between browser and note apps creates cognitive overload

3. **No Personalization**
   One-size-fits-all delivery ignores individual learning styles and accessibility needs

4. **Time Barriers**
   Manual transcription wastes valuable learning time

5. **Neurodiversity Gap**
   15–20% of learners with different cognitive needs lack appropriate support

Business Impact: Extended onboarding · Reduced productivity · Skills gaps · Wasted training investments

THE PROBLEM

Learners Are Overwhelmed

Fragmented tools, competing tabs, and no clear path create cognitive overload — making it harder to learn and retain knowledge.

- Too many tabs & distractions
- No AI-assisted study structure
- Wasted time, low retention

THE SOLUTION

Knowledge Wave

Seamlessly integrating AI-powered note-taking directly into Microsoft Learn

Knowledge Wave deploys intelligent assistance, real-time comprehension support, and personalized pathways that transform skill acquisition. Our adaptive agent system acts as an intelligent learning companion — understanding unique needs and providing contextual support precisely when needed.

Intelligent Assistance

AI-powered support that understands and responds to learner needs in real time

Adaptive Personalization

Dynamic pathways that adjust to individual learning styles and accessibility requirements

Real-Time Comprehension

Live analysis to assess understanding and surface targeted content on-demand

Combining cutting-edge AI with learning science and accessibility to reimagine corporate learning

TECHNICAL ARCHITECTURE

Built on Microsoft Azure

Enterprise Infrastructure

Azure OpenAI Service

Intelligent content generation and personalized learning recommendations

Azure Cognitive Services

Text analytics, sentiment analysis, and comprehension assessment

Azure Functions

Serverless real-time processing and event-driven workflows

Azure Blob Storage

Secure storage for user notes and all learning artifacts

Azure Key Vault

Enterprise-grade security for credentials and sensitive data

Azure Application Insights

Analytics, monitoring, and continuous optimization

INTELLIGENT FEATURES

Smart Widget Capabilities

Knowledge Wave transforms every keyword into a rich, interactive knowledge experience

01

Smart Widget

Auto-expands keywords into definitions, examples, and context-aware quick actions

02

Semantic Awareness

Detects themes and suggests related terms — understanding meaning, not just keywords

03

Knowledge Snippets

Every keyword becomes a rich mini-object with definitions, links, icons & version history

04

Drag & Drop Blocks

Tactile, modular note-building — drag knowledge blocks into documents visually

05

Smart Paste

Detects sentence structure and integrates keywords smoothly with suggested rewrites

06

Inline Search

Hover any keyword to instantly preview related notes, diagrams, and AI summaries

07

Auto Glossaries

Every inserted keyword auto-populates a living glossary and document index

08

Learning Mode

Quiz learners on keywords, track mastery, and generate flashcards automatically

Knowledge Wave

AI-GENERATED FEATURE

Auto-Generated AI Study Guide

Knowledge Wave uses AI to automatically create personalised study guides, task lists, and key term summaries — so learners can focus on understanding, not organizing.

Reading Passage

Principles of Complex Systems

AI surfaces the core reading material for the topic — formatted for clarity, with key concepts highlighted for quick navigation.

Task List

Auto-generated checklist

Review Chapter Introduction · Complete Self-Assessment Quiz · Formulate Discussion Questions · Find related articles

Key Study Terms

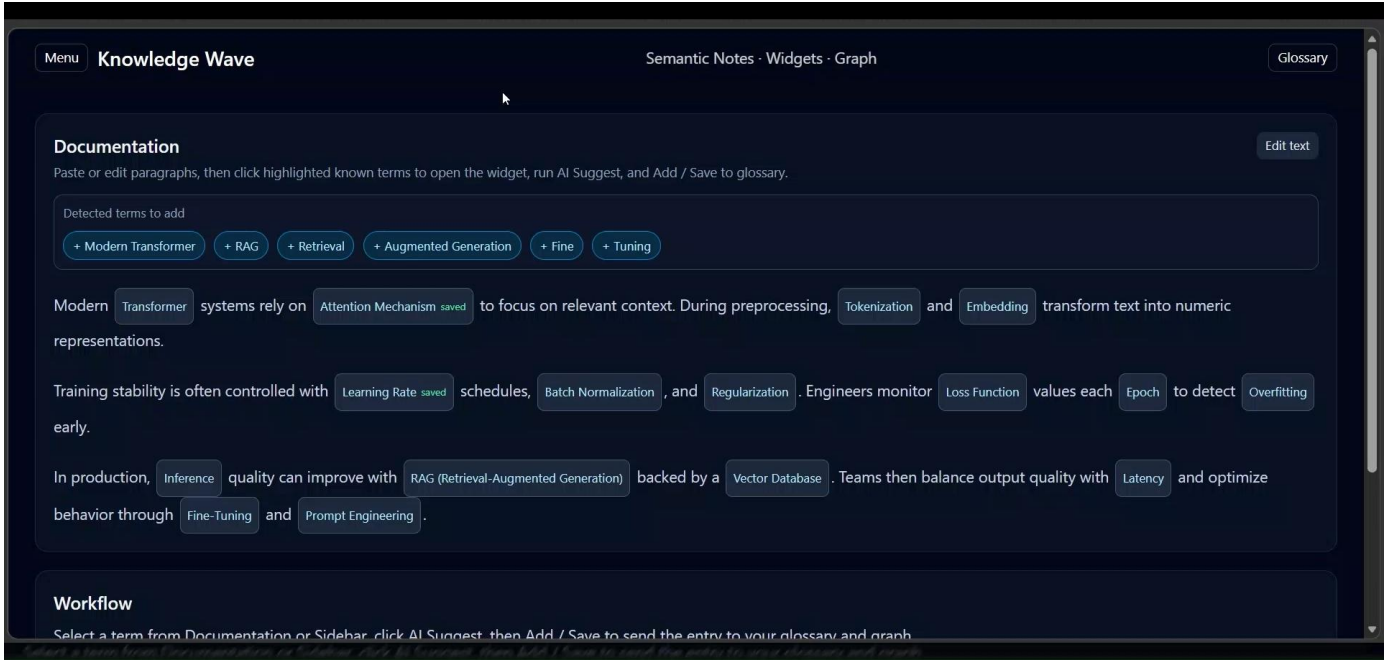
AI-curated vocabulary

Core Principles · feedback loop · emergence · non-linear dynamics · adaptive · agent-based model

AI generates personalised study content in real time — no manual prep required by learners or educators.

Built on Microsoft Azure

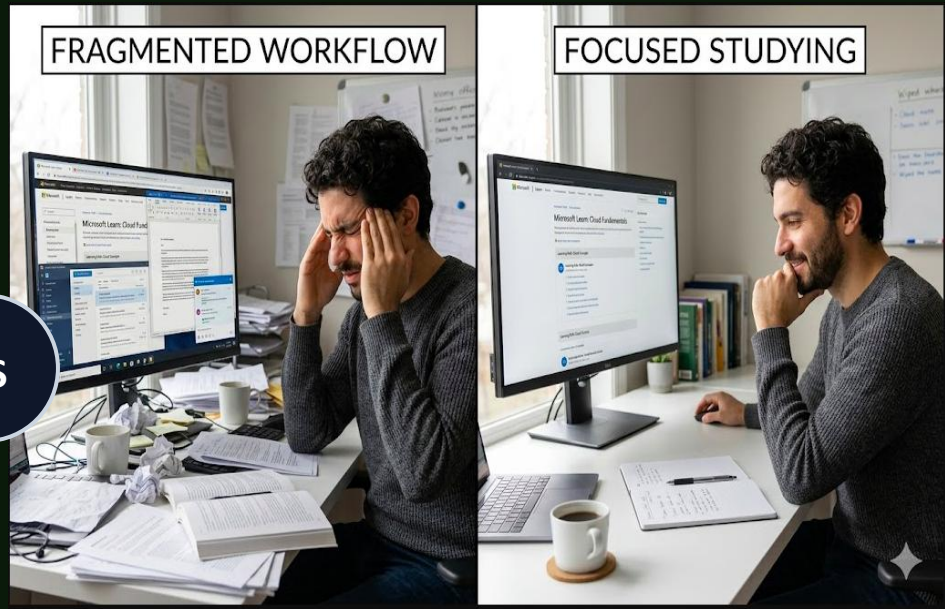
Enterprise Infrastructure



FRAGMENTED WORKFLOW

- Multiple tabs, apps & distractions
- No clear learning structure
- Cognitive overload & frustration

FOCUSED STUDYING



VS

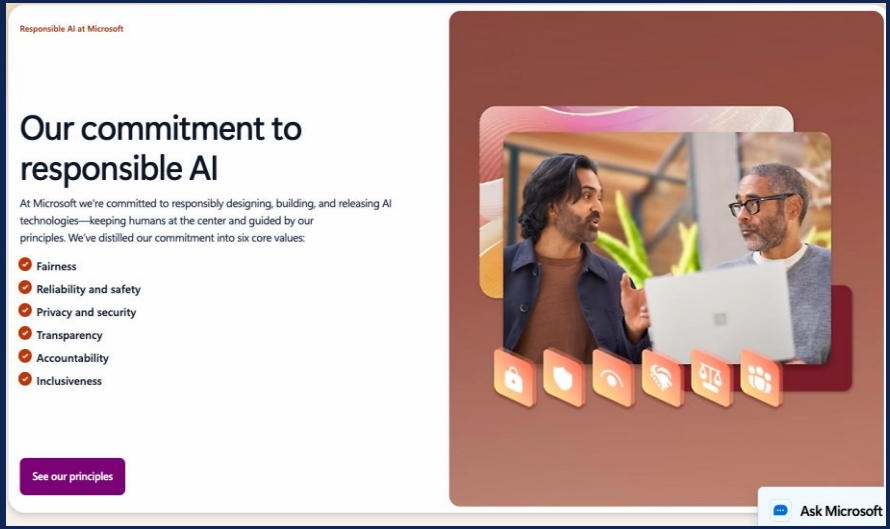
- Single focused learning environment
- AI-generated study guide & task list
- Clear progress & structured outcomes

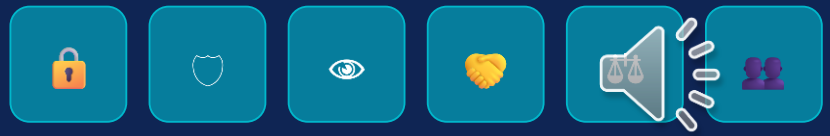
Responsible AI at Microsoft

Our commitment to Responsible AI

At Microsoft we're committed to responsibly designing, building, and releasing AI technologies — keeping humans at the center and guided by our principles. We've distilled our commitment into six core values:

- Fairness
- Reliability and safety
- Privacy and security
- Transparency
- Accountability
- Inclusiveness





EXPECTED OUTCOMES

Business Impact

Knowledge Wave transforms training from obligation into opportunity — building a true culture of continuous learning.

From onboarding to mastery,
every learner succeeds.

Five Key Outcomes

1. **Higher Completion Rates**
   Personalized engagement keeps learners on track and motivated to finish

2. **Faster Skill Application**
   Contextual, just-in-time learning accelerates time-to-competency

3. **Reduced Onboarding Time**
   Adaptive pathways cut ramp-up time with targeted content delivery

4. **Remote Workforce Engagement**
   Enhanced peer collaboration and community features bridge distance gaps

5. **Culture of Continuous Learning**
   Training transforms from a box to check into a habit employees value

Knowledge Wave

ROADMAP TIMELINE

PROTOTYPE → **PILOT** → **SCALE** → **CONTINUOUS IMPROVEMENT**

PROTOTYPE	PILOT	SCALE	CONTINUOUS IMPROVEMENT
≈	≈	≈	≈
Define & prototype concept solutions	Test with a small target group	Roll out to broader audience	Measure, iterate & keep improving

Green stages = currently active · Purple stages = planned · ≈ = Knowledge Wave waypoints

WIC AI INNOVATION CHALLENGE COMPETITION

Transforming Corporate Learning

Knowledge Wave partners with every learner to accelerate growth — combining cutting-edge AI, learning science, and accessibility to reimagine how organizations build capability.

23
Languages

15–20%
Neurodiverse Learners

6
Azure Services

+
Possibilities

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